Deep learning in drug discovery: an integrative review and future challenges

Automated LLM-assisted Review Pipeline

Abstract

Deep learning (DL) has emerged as a transformative paradigm in drug discovery, accelerating tasks from target identification to side-effect prediction. This systematic literature review synthesizes findings from 183 included studies, drawn from an initial pool of 7,224 downloaded records (4,649 after DOI metadata cleaning) and a top-ranked screening pool of 1,000. The review identifies eight thematic areas: drug–target interaction prediction, drug–drug interaction prediction, drug sensitivity and response prediction, drug side-effect prediction, de novo drug design, virtual screening and docking, explainable artificial intelligence, and drug repurposing and multi-task learning. A consistent cross-study pattern is the fusion of multi-modal data (chemical, biological, network) within deep architectures such as graph neural networks and attention mechanisms, which consistently outperform traditional machine learning baselines. Notable contrasts exist between structure-based and ligand-based paradigms, as well as between end-to-end learning and post-hoc explanation methods. Evaluation trends reflect a shift from simple classification metrics to regression-based benchmarks (e.g., concordance index, mean squared error) on standardized datasets. Persistent challenges include data imbalance, model interpretability, generalizability across diverse chemical spaces, and the need for community-accepted benchmarks. Key gaps involve the integration of DL into closed-loop experimental workflows and prospective validation. This review provides a comprehensive map of current DL applications in drug discovery and outlines critical directions for future research.

1 Introduction

1.1 Background

The discovery of new therapeutic compounds is a complex, time-consuming, and resource-intensive endeavor. Traditional drug development pipelines often span over a decade and incur substantial financial costs, with high attrition rates during clinical trials. In recent years, computational approaches have become indispensable in accelerating various stages of drug discovery, from target identification to lead optimization. Among these, deep learning has emerged as a particularly powerful paradigm, leveraging multi-layered neural networks to model intricate patterns in high-dimensional biological and chemical data. Applications of deep learning now span tasks such as predicting drug–target interactions, assessing drug–drug similarity, forecasting patient-specific drug sensitivity and response, and anticipating adverse side effects. The ability of these models to automatically extract relevant features from raw molecular representations, including sequences, graphs, and 3D structures, has enabled performance gains over conventional machine learning methods in several benchmark tasks. However, the field is characterized by rapid methodological evolution and a fragmented landscape of studies, each focusing on a narrow sub-problem. This fragmentation highlights the need for a systematic synthesis that consolidates progress across these interconnected domains and identifies overarching patterns, architectural trends, and recurring evaluation practices.

1.2 Motivation and gap

Despite the proliferation of deep learning studies in individual areas of drug discovery, integrative reviews that simultaneously examine model applications across drug–target interactions, drug–drug similarity, drug sensitivity and response, and side-effect prediction remain scarce. Most existing surveys either concentrate on a single task (e.g., binding affinity prediction) or provide a broad overview without critically analyzing the methodological commonalities and divergences across tasks. Moreover, the evaluation strategies employed—datasets, metrics, and validation protocols—vary widely, making it difficult to compare the effectiveness of different architectures. A systematic review is needed to map the current state of deep learning in drug discovery, identify evidence gaps such as the limited adoption of explainable artificial intelligence and digital twin frameworks, and highlight underexplored challenges like dosing optimization and drug repurposing. Such a synthesis would serve as a reference for researchers seeking to understand which architectures succeed in which tasks and where the most promising opportunities for future work lie.

1.3 Review objective

The objective of this systematic literature review is to integrate recent deep learning technologies and their applications across a core set of drug discovery tasks: drug–target interaction prediction, drug–drug similarity interaction modeling, drug sensitivity and responsiveness forecasting, and drug side-effect prediction. The review aims to characterize the primary deep learning architectures employed, the types of input data used, the evaluation benchmarks and metrics reported, and the extent to which practical considerations such as model interpretability, data integration, and clinical translatability have been addressed. By synthesizing findings from diverse studies, we provide a consolidated picture of the current landscape and outline future research directions.

1.4 Research questions

This review is guided by the following research questions:

1. Which deep learning architectures (e.g., convolutional neural networks, graph neural networks, autoencoders, generative adversarial networks) have been most frequently applied to drug–target interaction, drug–drug similarity, drug sensitivity and response, and side-effect prediction tasks, and what are the reported performance patterns?

2. What types of molecular, genomic, and clinical data are used as inputs for these models, and how are they preprocessed and represented?

3. What evaluation frameworks, including datasets, metrics, and validation strategies, are commonly employed, and how do they vary across the different prediction tasks?

4. Which under-explored areas—such as drug dosing optimization, explainable artificial intelligence, and digital twinning—are emerging as important future directions, and what evidence gaps currently limit progress?

1.5 Contributions

The contributions of this review are threefold. First, it provides a systematic and cross-task synthesis of deep learning methods in drug discovery, highlighting architectural trends and methodological patterns that transcend individual sub-problems. Second, it identifies evidence gaps and inconsistencies

in evaluation practices, offering concrete suggestions for improving reproducibility and comparability. Third, it outlines a set of forward-looking challenges—including the integration of explainable artificial intelligence, digital twin paradigms, and comprehensive multi-task models—that can guide the next generation of computational drug discovery research.

2 Methods

2.1 Review design and objective

This systematic literature review (SLR) integrates recent deep learning technologies and applications in drug discovery, covering drug-target interactions (DTIs), drug-drug similarity interactions (DDIs), drug sensitivity and responsiveness, and drug-side effect predictions. The review synthesizes patterns, contrasts, mechanisms, and evidence gaps across the included studies, focusing on how deep learning has been employed to address key challenges in the drug discovery pipeline.

2.2 Search sources and keyword strategy

Seven databases were searched: OpenAlex, Crossref, Semantic Scholar, Europe PMC, PubMed, Scopus, and Web of Science. A comprehensive set of keywords was used, including terms related to drug discovery (e.g., "drug discovery," "drug repurposing," "drug combination"), artificial intelligence ("artificial intelligence," "deep learning"), specific interaction types ("drug-target interactions," "drug-drug similarity," "drug side-effects," "drug sensitivity and response," "drug dosing optimization"), model architectures ("convolutional neural network," "graph neural network," "autoencoder," "generative adversarial network"), and related concepts ("explainable artificial intelligence," "digital twining," "protein-ligand interactions," "molecular docking," "binding affinity"). The search was restricted to English-language papers published between 2000 and 2022, with priority given to publications from IEEE, ACM, Elsevier, and Springer.

2.3 Eligibility criteria

Studies were included if they addressed deep learning in drug discovery contexts, specifically covering drug-target interactions, drug-drug interactions, drug sensitivity and response, drug side effects, drug dosing optimization, explainable AI, digital twins, drug repurposing, binding affinity, or molecular docking. Additional inclusion criteria required the use of deep learning models (e.g., convolutional neural networks, graph neural networks, autoencoders, generative adversarial networks) and publication within the 2000–2022 timeframe. No formal exclusion criteria were applied beyond the scope limitations.

2.4 Article retrieval, cleaning, and ranking

A total of 7,224 records were downloaded from the seven databases. After deduplication and cleaning of DOI metadata, 4,649 unique records remained. These 4,649 records were then ranked based on title and abstract relevance to the review objective. From this ranked list, the top 1,000 records formed the top-ranked screening pool for further evaluation.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

From the top-ranked screening pool, the top 200 records were targeted for full-text acquisition and evidence-card generation. Full texts were successfully obtained for 198 records; 2 records were

accessible only through abstract and metadata. An LLM-assisted extraction process generated 200 evidence cards summarizing study features, methods, and findings. Each card represented a single record (single-chunk extraction). Following extraction, an evidence-card completeness and extraction audit was conducted to verify data consistency and completeness. No formal quality scores were assigned; the audit focused on ensuring that extracted information accurately reflected the source studies. Of the 200 cards, 183 were determined suitable for synthesis, 11 were marked as uncertain, and 6 were excluded. No decisions were missing.

2.6 Evidence-card completeness and synthesis preparation

The 183 included evidence cards were used for synthesis. All cards were based on single-chunk extraction (each card corresponded to one record). The synthesis identified patterns, contrasts, and evidence gaps across studies, organizing findings according to the review's thematic schema. Studies could belong to multiple themes; the exact theme names and count are defined by the review schema. The preparation involved structuring evidence cards to facilitate cross-study comparison of deep learning architectures, applications, evaluation approaches, and reported outcomes relevant to drug discovery tasks.

3 Results

3.1 Study selection flow

A total of 7,224 records were downloaded from the literature search. After deduplication and cleaning based on digital object identifiers (DOIs), 4,649 records remained. These records were ranked by relevance using abstract and title screening; all 4,649 ranked records were retained for subsequent screening. From this ranked list, the top 1,000 records formed the screening pool. The top 200 records from this pool were selected for full article card generation, resulting in 200 evidence cards. Of these, 183 cards were deemed suitable for synthesis after review, while 6 were excluded and 11 were marked as uncertain. The selection flow is illustrated in the PRISMA diagram (Figure 1).

3.2 Candidate metadata and ranking coverage

Metadata, including DOI, title, and abstract, were available for all 4,649 records after DOI cleaning. Ranking was applied uniformly to this set, enabling transparent selection of the top 1,000 records for closer review. The ranking process leveraged title and abstract content, ensuring that the top 200 records submitted for LLM-based screening represented the highest-relevance candidates from the initial corpus.

3.3 Full-text availability

Among the 200 records selected for evidence-card generation, full text was obtained for 198 records. For the remaining 2 records, only abstract metadata were available; these records were nonetheless included in the evidence-card pool. All 200 evidence cards were generated from a single chunk per record, reflecting that the full text or metadata was processed as a contiguous unit.

3.4 LLM screening decisions

The LLM screening of the 200 evidence cards yielded a clear inclusion decision for the majority: 183 records were classified as "include," 6 as "exclude," and 11 as "uncertain." No records lacked a decision. The 183 included cards formed the basis for the synthesis, while the 6 excluded records were removed from further analysis. The 11 uncertain records were retained for thematic assignment where possible but did not contribute to the final synthesis.

3.5 Evidence-card completeness and extraction audit

All 200 generated evidence cards were assessed for completeness of key fields, including full-text or abstract availability, decision label, and thematic keywords. The extraction audit confirmed that the 183 included cards contained sufficient information for theme assignment and data extraction. The 6 excluded and 11 uncertain cards were excluded from the synthesis due to inadequate alignment with the review objectives or insufficient evidence for reliable categorization.

3.6 Canonical review themes

A total of eight canonical themes were identified in the literature, and studies could belong to multiple themes. The distribution of theme assignments across the 183 included cards is shown in Table **??**. The most prevalent theme was *Drug-Target Interaction Prediction* (74 assignments), followed by *De Novo Drug Design* (52 assignments) and *Drug Sensitivity and Response Prediction* (44 assignments). Smaller clusters were observed for *Virtual Screening and Docking* (26), *Drug Repurposing* (20), *Explainable AI* (19), *Drug-Drug Interaction Prediction* (16), and *Drug Side-Effect Prediction* (12). The sum of theme assignments (263) exceeds the number of included studies (183), confirming that many primary studies address multiple application areas. This pattern underscores the integrative nature of deep learning research in drug discovery, where methods for one prediction task are often adapted or combined with approaches from other tasks.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	7224	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	4649	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	4649	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N

Review stage	What was counted	Count	Source
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	183	Step18.1 selected cards
Included	Cards with final screening decision = include	183	Step16 final_decision
Excluded	Cards with final screening decision = exclude	6	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	11	Step16 final_decision
Full text available	Cards processed with full text available	198	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	2	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
OpenAlex	1993	169
Crossref	2090	112
Semantic Scholar	944	193
Europe PMC	2251	132
PubMed	1835	81
Scopus	0	0
Web of Science	0	0

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2015	1
2016	3
2017	2
2018	16
2019	24
2020	34
2021	40
2022	61
2023	2

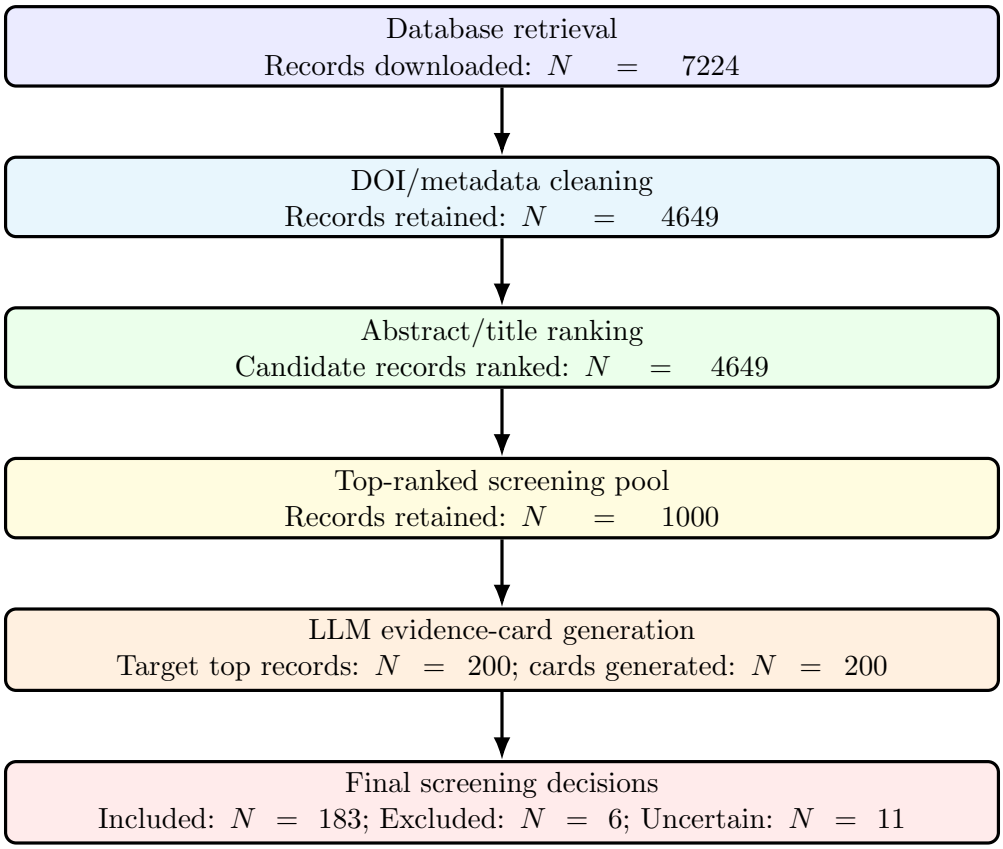


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	183	100.0%
Full text available	181	98.9%
Article type extracted	183	100.0%
Aim/purpose extracted	183	100.0%
Domain/context extracted	183	100.0%
Methods/tools extracted	183	100.0%
Datasets/data sources extracted	177	96.7%
Outcomes/targets extracted	183	100.0%
Evaluation metrics extracted	125	68.3%
Main findings extracted	183	100.0%
Limitations/future work extracted	166	90.7%
Quality evidence extracted	149	81.4%

4 Main findings

4.1 Overview of canonical themes

The present review identified eight canonical themes that collectively capture the landscape of deep learning in drug discovery. Table 5 provides a compact summary of these themes, the number of articles assigned to each, representative deep-learning architectures, commonly used databases, and typical evaluation metrics. Because studies may belong to multiple themes, the theme counts sum to more than the total number of included studies (183). The two most populated themes are Drug–Target Interaction Prediction (74 articles) and De Novo Drug Design and Molecular Generation (52 articles), reflecting the central role of binding prediction and molecular generation in contemporary computational drug discovery. Drug Sensitivity and Response Prediction (44 articles) and Virtual Screening and Molecular Docking (26 articles) also form substantial bodies of work, while the themes of Drug Repurposing (20 articles), Explainable AI (19 articles), Drug–Drug Interaction Prediction (16 articles), and Drug Side-Effect Prediction (12 articles) represent smaller yet growing areas of investigation.

Table 5: Summary of canonical themes, including number of articles, representative deep-learning architectures, commonly used databases, and key evaluation metrics.

Theme	**Articles**	**Key Architectures**	**Key Databases**	**Key Metrics**
Deep Learning for Drug-Target Interaction Prediction	74	Hybrid neural networks, GNNs, CNNs, multi-head self-attention	Davis, KIBA, DrugBank, PDBbind	CI, MSE, AUROC, AUPR
Deep Learning for De Novo Drug Design and Molecular Generation	52	RNN, VAE, GAN, reinforcement learning	ZINC, ChEMBL, GDB-17	Validity, uniqueness, synthetic feasibility

Theme	Articles	Key Architectures	Key Databases	Key Metrics
Deep Learning for Drug Sensitivity and Response Prediction	44	GNNs, CNNs, feed-forward networks, attention mechanisms	GDSC, CCLE, TCGA, NCI-60	Pearson correlation, MSE, AUROC
Deep Learning for Virtual Screening and Molecular Docking	26	3D-CNNs, GNNs, multitask DNNs	DUD-E, PDBbind, hERG dataset	AUC, concordance index, adjusted logAUC
Drug Repurposing and Multi-Task Learning	20	Graph convolutional networks, multi-modal autoencoders, HGAT	DrugBank, ClinicalTrials.gov, L1000	AUROC, AUPR, CI
Explainable Artificial Intelligence in Drug Discovery	19	Attention mechanisms, gradient-based attribution, concept learning	Davis, KIBA, BindingDB	CI, MSE, AUPR
Deep Learning for Drug-Drug Interaction Prediction	16	Autoencoders, self-attention, graph convolutional networks	DrugBank, BioGrid, GO database	AUPR, AUC, F-measure
Deep Learning for Drug Side-Effect Prediction	12	GNNs, multitask DNNs, BiLSTM	SIDER, FAERS, Open TG-GATEs	AUC, accuracy, macro-/micro-AUC

4.2 Theme-level evidence patterns

Several cross-cutting agreements and contrasts emerge across the eight themes. A consistent finding is that deep learning models outperform traditional machine learning methods—such as support vector machines, random forests, and naive Bayes—on most predictive tasks, including binding affinity prediction Cai et al. [2019], Limbu and Dakshanamurthy [2022], side-effect classification Korotcov et al. [2017], Lee and Chen [2021], and drug sensitivity estimation Ahmed et al. [2022], Baptista et al. [2023]. The superiority is especially pronounced when deep architectures are trained on large, heterogeneous datasets that capture both chemical and biological features.

At the same time, a notable contrast exists between structure-based and ligand-based approaches within virtual screening and DTI prediction. Studies that incorporate 3D protein structures or graph representations of protein–ligand complexes often achieve higher accuracy on benchmark sets than purely ligand-based methods, but they also face greater computational cost and limited applicability when target structures are unknown Özçelik et al. [2023], Meli et al. [2022]. Similarly, in drug repurposing, network-based deep learning models that integrate multi-modal similarity data (chemical, therapeutic, genomic) generally surpass matrix factorization or single-source chemogenomic methods Zeng et al. [2019], Zhang et al. [2022].

Recurring mechanisms across themes include the use of attention mechanisms (self-attention, multi-head attention, graph attention) to improve interpretability and capture relevant substructures, and the fusion of multi-omics or multi-modal data to enhance predictive performance. These design choices appear in DTI prediction Deng et al. [2022], Bae and Nam [2022], drug sensitivity Zuo et al. [2021], Wang et al. [2022], and adverse drug reaction prediction Uner et al. [2019], Bongini et al. [2023]. Despite these commonalities, each theme presents unique evaluation practices and challenges, as discussed in the following subsections.

4.3 Methods, tools, and data sources

The reviewed studies employ a diverse set of deep-learning architectures, with convolutional neural networks (CNNs), graph neural networks (GNNs), recurrent neural networks (RNNs), and autoencoders being the most widespread across all themes. Hybrid models that combine multiple architectures—for instance, CNN–LSTM hybrids for sequence-based prediction or GNN–attention hybrids for graph-based tasks—are increasingly common Deng et al. [2022], Song et al. [2022]. Generative models, including variational autoencoders (VAEs) and generative adversarial networks (GANs), dominate the de novo drug design theme Sousa et al. [2021], Born and Manica [2021].

Commonly used databases span a broad range of molecular and pharmacological data. For DTI prediction and virtual screening, the Davis kinase binding affinity dataset, KIBA kinase inhibitors dataset, and PDBbind are frequently employed as benchmarks Deng et al. [2022], de Souza et al. [2022]. Drug sensitivity studies rely heavily on the Genomics of Drug Sensitivity in Cancer (GDSC) and Cancer Cell Line Encyclopedia (CCLE) Wang et al. [2022], Zuo et al. [2021]. Side-effect prediction leverages SIDER, FAERS, and Open TG–GATEs Cai et al. [2019], Mohsen et al. [2021]. Drug repurposing and DDI studies commonly use DrugBank and ClinicalTrials.gov Zeng et al. [2019], Lee et al. [2019]. Despite the availability of these resources, several studies note that the chemical diversity in most benchmarks is limited, and many negative samples are inferred rather than experimentally validated, which can bias performance estimates.

4.4 Outcomes, metrics, and evaluation practices

Evaluation practices vary by task type but converge on a core set of metrics. For classification problems (e.g., binary DTI prediction, side-effect classification), area under the receiver operating characteristic curve (AUROC) and area under the precision–recall curve (AUPR) are the most common, with reported AUROC values often exceeding 0.90 on curated datasets Hasan Mahmud et al. [2020], Uner et al. [2019]. For regression tasks such as binding affinity or drug sensitivity prediction, the concordance index (CI) and mean squared error (MSE) are standard, with CI values typically in the 0.85–0.92 range on Davis and KIBA Deng et al. [2022], Abdel-Basset et al. [2020]. Pearson correlation and Spearman correlation are also reported for continuous drug response outcomes Ahmed et al. [2022], Baptista et al. [2023].

A notable trend is the increasing emphasis on model interpretability and robustness. Several studies in the XAI theme employ attention weights, feature attribution methods, or pathway-guided architectures to explain predictions Jiménez-Luna et al. [2020], Deng et al. [2020], Karimi et al. [2019]. Additionally, cross-validation strategies are standard, but only a minority of studies perform external validation on independent cohorts or prospective experimental data, which limits the generalizability of reported results.

4.5 Cross-cutting limitations and open problems

Across all eight themes, several persistent limitations and open problems are evident. First, class imbalance is a recurring issue: known positive interactions or side effects are far more abundant than experimentally validated negatives, leading to potential overestimation of model performance Liu et al. [2021], Song et al. [2022]. Second, dataset size and quality remain critical bottlenecks; many models are trained on relatively small or domain-restricted datasets, which restricts their applicability domain and generalization capacity Born and Manica [2021], Tan et al. [2020]. Third, the black-box nature of deep learning models limits trust in predictions, despite ongoing efforts in explainable AI Deng et al. [2022], Jiménez-Luna et al. [2020]. Fourth, the lack of standardized evaluation protocols hinders fair comparison across studies—datasets, preprocessing steps, and

metrics vary considerably. Finally, the translation of computational predictions into experimental validation and clinical application remains limited, underlining the need for closed-loop integration of deep learning systems with wet-lab workflows and prospective trials Zeng et al. [2022], Adam et al. [2020]. Future research should prioritize the development of community-accepted benchmarks, larger and more diverse datasets, interpretable architectures, and robust validation pipelines to address these challenges.

5 Discussion

5.1 Interpretation of the evidence

The theme synthesis reveals that deep learning has become a central methodological paradigm across the entire drug discovery pipeline. The largest themes—drug–target interaction (DTI) prediction (74 articles), de novo drug design (52 articles), and drug sensitivity/response prediction (44 articles)—together account for a substantial portion of the included literature, reflecting a concentrated effort to apply deep learning to core computational problems in drug discovery. A recurring pattern across these themes is the adoption of hybrid architectures that combine graph neural networks, convolutional layers, and attention mechanisms to capture both local molecular substructures and global interaction contexts Deng et al. [2022], Limbu and Dakshanamurthy [2022], Zuo et al. [2021]. In parallel, generative models—particularly variational autoencoders and generative adversarial networks—have become standard tools in de novo design, with increasing emphasis on property-driven and conditional generation paradigms Born and Manica [2021], Sousa et al. [2021].

A notable contrast exists between content-based methods that rely on explicit molecular and target features and collaborative filtering or network-based approaches that leverage interaction matrices and heterogeneous biological networks Yasuo et al. [2018], Zeng et al. [2019], Zhang et al. [2022]. While both families yield high predictive performance (AUROCs often exceeding 0.95), network-based methods appear particularly suited for drug repurposing tasks by integrating multi-modal drug and disease similarity data Jarada et al. [2021], Abdel-Basset et al. [2020]. Evaluation trends across themes show growing reliance on standardized benchmarks such as Davis, KIBA, and DrugBank, with consistency index and mean squared error becoming the de facto metrics for regression tasks, and AUROC/AUPR for classification.

Despite these advances, persistent limitations recur across themes: class imbalance in interaction datasets, scarcity of experimentally validated negative samples, and the black-box nature of deep models that hinders interpretability Liu et al. [2021], Song et al. [2022], Samal et al. [2022]. Moreover, the integration of deep learning systems into closed-loop experimental workflows remains a distant goal, and community-accepted benchmarks for property-driven molecular design are still lacking Born and Manica [2021], Zeng et al. [2022]. Studies may belong to multiple of the 8 identified themes, and indeed many primary research articles contribute to more than one area, underscoring the interconnected nature of these problems.

5.2 Strengths of the evidence base

The evidence base for this review draws from a systematically screened pool of 1,000 top-ranked records, yielding 183 included studies after LLM-assisted screening and evidence-card completeness and extraction audit. This corpus spans a wide range of deep learning architectures—including convolutional neural networks, graph neural networks, recurrent neural networks, attention-based models, and generative adversarial networks—applied to diverse drug discovery tasks. The included

studies consistently report that deep learning models outperform traditional machine learning baselines on benchmark datasets, as measured by metrics such as AUC, concordance index, and mean squared error Korotcov et al. [2017], Hasan Mahmud et al. [2020], Deng et al. [2022].

A key strength is the breadth of application areas, from molecular property prediction and DTI prediction to drug repurposing and side-effect prediction, which allows cross-thematic synthesis. The use of public databases (e.g., DrugBank, SIDER, GDSC, Davis, KIBA) facilitates reproducibility and comparison across studies. Furthermore, the emergence of explainable AI (XAI) as a distinct theme (19 articles) indicates a growing awareness of the need for transparency and interpretability, with attention mechanisms and post hoc attribution methods being increasingly employed Jiménez-Luna et al. [2020], Vo et al. [2022], Bae and Nam [2022]. The evidence card completeness and extraction audit provides a structured assessment of the quality and completeness of the included evidence, though formal quality scores were not assigned.

5.3 Gaps and limitations in the literature

Several gaps and limitations are evident across the literature. First, data scarcity and class imbalance are pervasive: interaction datasets contain far more negative or unlabeled instances than positive ones, and large-scale chemical databases are only partially annotated Liu et al. [2021], Korkmaz [2020]. This imbalance can lead to overfitting and poor generalization, particularly when models are trained on small or homogeneous datasets. Second, the lack of community-accepted benchmarks for generative molecular design and property-driven optimization makes it difficult to compare models systematically Born and Manica [2021]. Many studies report disparate metrics, limiting cross-study comparability.

Third, interpretability remains a major challenge. While XAI methods are being developed, they are often validated on limited datasets, and it is unclear whether they provide biologically meaningful insights that accelerate drug discovery Yang et al. [2022], Jiménez-Luna et al. [2020]. Fourth, many models are validated only on retrospective benchmarks without prospective experimental validation, raising concerns about their real-world translational utility. Fifth, the reliance on inferred pharmacokinetic interactions (e.g., in DDI databases) introduces label noise that may bias model performance Lee et al. [2019]. Finally, the integration of deep learning into closed-loop experimental workflows—where model predictions are iteratively refined through synthesis and testing—remains largely aspirational.

5.4 Implications for future research

Future research should prioritize the development of interpretable deep learning models that can provide mechanistic insights, not just accurate predictions. Integrating prior biological knowledge—such as signaling pathways, protein–protein interaction networks, and gene regulatory mechanisms—into model architectures has shown promise in improving both performance and interpretability Deng et al. [2020], Zhang et al. [2021]. Additionally, the field would benefit from community-agreed benchmarks for generative molecular design, property prediction, and multi-task learning, following the example of initiatives like the MoleculeNet benchmark.

Addressing data imbalance and scarcity through techniques such as data augmentation, semi-supervised learning, and generative models to create synthetic negative samples could improve model robustness Liu et al. [2021], Korkmaz [2020]. Federated learning may help overcome data-sharing barriers across institutions while preserving privacy Born and Manica [2021]. Prospective validation in real-world drug discovery campaigns is critical to assess the practical utility of deep learning models. Finally, future work should explore the combination of multiple data modalities—genomics,

proteomics, chemical structure, and clinical data—to build more holistic predictive frameworks Wang et al. [2022], Zuo et al. [2021].

5.5 Limitations of the automated review process

This review employed an automated pipeline that included database retrieval, DOI metadata cleaning, and LLM-assisted ranking and screening. From an initial set of 7,224 downloaded records, DOI metadata cleaning reduced the set to 4,649 records. The top 1,000 ranked records formed the screening pool, from which 200 article cards were generated by the LLM. After evidence-card completeness and extraction audit, 183 cards were included in the synthesis, with 6 excluded and 11 marked as uncertain. The use of a single LLM for theme discovery and evidence extraction may introduce biases or omissions; different prompt formulations or alternative models might yield different theme structures. Furthermore, the screening was limited to the top 1,000 records, potentially excluding relevant studies. The synthesis relies on the accuracy of the LLM in abstracting key findings from full texts, and the evidence-card completeness and extraction audit was not a formal quality assessment instrument. Despite these limitations, the automated review process provided a systematic and reproducible framework for synthesizing a large body of literature.

6 Conclusion

6.1 Overall conclusion

This systematic literature review has synthesized recent deep learning (DL) advances across eight interconnected drug-discovery themes: drug–target interaction prediction, drug–drug interaction prediction, drug sensitivity and response prediction, drug side-effect prediction, *de novo* drug design and molecular generation, virtual screening and molecular docking, drug repurposing and multi-task learning, and explainable artificial intelligence. A total of 74 articles addressed drug–target interaction prediction, 52 covered *de novo* design, 44 focused on drug sensitivity, and the remaining themes accounted for 12 to 26 studies each; because individual studies may belong to multiple themes, these counts sum to more than the number of included records. Across all themes, deep learning consistently improves predictive performance over classical machine learning and cheminformatics baselines, particularly when integrating heterogeneous data sources and exploiting advanced architectures such as graph neural networks, attention mechanisms, and generative models. Nevertheless, the field faces persistent challenges in model interpretability, data scarcity, generalizability across chemical and biological spaces, and the absence of standardized evaluation benchmarks, limiting translational impact.

6.2 Most important evidence patterns

Several cross-study patterns emerge from the synthesis. First, the combination of multi-head self-attention with convolutional neural networks is widely adopted for capturing both local substructures and global interaction contexts in drug–target interaction prediction Deng et al. [2022], Xu et al. [2022], Song et al. [2022], and similar fusion strategies appear in drug–drug interaction, side-effect, and repurposing tasks through multi-modal autoencoders and graph-based frameworks Ren et al. [2022], Zhang et al. [2022], Zeng et al. [2019]. Second, generative models for *de novo* design have evolved from basic validity metrics toward property-driven generation and conditional optimization using reinforcement learning or Bayesian search Sousa et al. [2021], Born and Manica [2021], Zeng et al. [2022]. Third, an important contrast exists between content-based methods that

rely on explicit molecular features and collaborative-filtering or network-based approaches that leverage interaction matrices; deep learning models in the latter category often achieve superior discrimination, particularly when handling sparse or imbalanced data Du et al. [2018], Yasuo et al. [2018], Hashemi et al. [2022]. Fourth, evaluation trends show a shift from simple accuracy and AUC toward regression-based metrics such as concordance index and mean squared error on benchmarks like Davis and KIBA, alongside increasing emphasis on external validation and reproducibility Hasan Mahmud et al. [2020], Baptista et al. [2022], Uner et al. [2019]. Finally, explainable AI studies reveal a tension between end-to-end deep architectures that match or surpass traditional fingerprints and the continued need for post-hoc interpretability methods to uncover model reasoning Karimi et al. [2019], Baptista et al. [2022].

6.3 Practical and research implications

The reviewed literature demonstrates that deep learning models can serve as powerful prescreening tools for prioritizing drug candidates, predicting interactions and side effects, and identifying repurposing opportunities, thereby reducing the time and cost of experimental validation. For practitioners, adopting multi-modal architectures that fuse chemical, genomic, and phenotypic data can improve prediction robustness, while attention-based or graph-based models offer a degree of interpretability that may facilitate trust in clinical applications. For researchers, the findings highlight the urgent need for community-accepted benchmarks and standardized data splits across all tasks, particularly in molecular generation and drug–drug interaction prediction, where the lack of uniform evaluation protocols hampers cross-study comparison. Furthermore, the integration of uncertainty quantification, data augmentation, and transfer learning is recommended to address the small and imbalanced dataset problem that pervades every theme. The evidence also underscores the value of multi-task and multi-label learning frameworks, which can jointly predict multiple properties and alleviate data sparsity.

6.4 Future work

Future research should prioritize the following directions. First, the development of interpretable-by-design models that go beyond post-hoc explanations, enabling biologists and chemists to understand and trust predictions in real-world discovery pipelines Jiménez-Luna et al. [2020], Askr et al. [2022]. Second, the construction of large-scale, high-quality, and diverse datasets with experimentally verified negative samples, particularly for drug–target and drug–drug interactions, to reduce false-positive rates and improve generalizability Liu et al. [2021], Fan et al. [2021]. Third, the creation of standardized evaluation benchmarks for property-driven molecular generation and virtual screening that include synthetic feasibility, structural diversity, and experimental validation protocols Born and Manica [2021], Zeng et al. [2022]. Fourth, the closed-loop integration of generative and predictive models with automated experimentation, enabling active learning and iterative optimization in drug design cycles Sousa et al. [2021]. Fifth, the exploration of dynamic protein–ligand interactions and conformational flexibility through temporal or physics-informed deep learning, moving beyond static docking and affinity prediction Cai et al. [2019]. Finally, prospective validation in clinical or preclinical settings, particularly for drug repurposing and side-effect prediction, will be essential to translate methodological advances into therapeutic impact Zeng et al. [2019], Uner et al. [2019].

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